# 1 Basic Category Theory

**Remark 1.1.** *The notion of "class" in ZFC requires a definition. This can be done ("parametrically definable classes"– e.g., talk about "sets satisfying a formula" as opposed to "a set of things satisfying said formula"), but for now let's blackbox the concept and just say "classes are a generalization of sets which are allowed to be 'large'", e.g. so a "class of all sets" makes sense even though a "set of all sets" does not. For this course, this shouldn't matter ultimately anyways.*[1]

**Definition 1.1.** *A **Category**, $\mathcal{C}$, is a triple of classes, $\mathsf{obj}(\mathcal{C})$, $\mathsf{Hom}(\mathcal{C})$, and "$\circ$", AKA $\mathsf{Comp}(\mathcal{C})$, such that:*

- *For every $a, b \in \mathsf{obj}(\mathcal{C})$, there is a set $\mathsf{Hom}_{\mathcal{C}}(a, b)$ of elements of $\mathsf{Hom}(\mathcal{C})$.*
- *For every $a, b, c \in \mathsf{obj}(\mathcal{C})$, composition is a "map" $\circ : \mathsf{Hom}_{\mathcal{C}}(a, b) \times \mathsf{Hom}_{\mathcal{C}}(b, c) \to \mathsf{Hom}_{\mathcal{C}}(a, c)$*
  - *Composition is associative, in that $(f \circ g) \circ h = f \circ (g \circ h)$ for any morphisms $f, g, h \in \mathsf{Hom}(\mathcal{C})$ (whenever this composition makes sense).*
- *For every $a \in \mathsf{obj}(\mathcal{C})$, there is an element $\mathrm{id}_a \in \mathsf{Hom}_{\mathcal{C}}(a, a)$ which is an identity for composition.*

In short: we have objects, arrows between objects, a way of composing arrows, and identity arrows on each object. Oftentimes, when working with a category of modules or vector spaces, we will omit the name of the full category and simply denote $\mathsf{Hom}_R(M, N)$ for (say) $\mathsf{Hom}_{{}_R\mathsf{Mod}}(M, N)$.
The next definition is often of little intrinsic import[2], but will be a convenient formal thing to have for discussing "contravariant functors", and more generally for giving a nice notion of "duality" for many arguments.

**Definition 1.2.** *Given a category $\mathcal{C}$, we define its opposite category $\mathcal{C}^{op}$ to be the category with the same objects, but such that for every morphism $f : a \to b$ in $\mathsf{Hom}(\mathcal{C})$ we have an arrow $f^{op} : b \to a$ in $\mathsf{Hom}(\mathcal{C}^{op})$. Composition is defined by saying that, for every*

$$a \xrightarrow{f} b \xrightarrow{g} c$$

*in $\mathsf{Hom}(\mathcal{C})$,*

$$c \xrightarrow{(g \circ f)^{op}} a := c \xrightarrow{g^{op}} b \xrightarrow{f^{op}} a$$

*in $\mathsf{Hom}(\mathcal{C}^{op})$.*

**Definition 1.3.** *Given two categories, $\mathcal{C}$ and $\mathcal{D}$, a **functor** $F : \mathcal{C} \to \mathcal{D}$ is a map on objects $F_{\mathsf{obj}} : \mathsf{obj}(\mathcal{C}) \to \mathsf{obj}(\mathcal{D})$ together with a collection of maps on morphisms $F_{\mathsf{Hom}(a,b)} : \mathsf{Hom}_{\mathcal{C}}(a, b) \to \mathsf{Hom}_{\mathcal{D}}(Fa, Fb)$ such that:*

- *$F$ respects composition, e.g. $F_{\mathsf{Hom}(b,c)}(\varphi) \circ F_{\mathsf{Hom}(a,b)}(\psi) = F_{\mathsf{Hom}(a,c)}(\varphi \circ \psi)$*
- *$F$ maps identities to identities, e.g. $F_{\mathsf{Hom}(a,a)}(\mathrm{id}_a) = \mathrm{id}_{F_{\mathsf{obj}}(a)}$*

*. By convention we omit the subscripts and, for any $\mathcal{C}$-object $a$ or any $\mathcal{C}$-morphism $\varphi$, simply denote $Fa$ and $F\varphi$ for the corresponding object/morphism in $\mathcal{D}$.*

You'll note: this is basically the definition of a group homomorphism, but we append "takes identities to identities" since this isn't a guarantee in the absense of inverses (which, in general, is just the definition of a "monoid homomorphism"– a map of groups with identity but without inverses). This is not mere coincidence, and is in some ways how you "should" think about it.[3]

[1] This starts to actually be worth noting [especially supposing you care about working in ZFC or even NBG without universes] once certain constructions/objects (e.g. derived categories, functor categories, stacks etc.) become directly relevant to you. That said, if you aren't a logician then I believe it *mostly* fine to blackbox such foundational issues until it comes up naturally.

[2] This is slightly illegitimate– there are indeed cases where a category of interest has, as its dual, another category of interest (up to equivalence). For algebraic geometers, one has the category of affine schemes being equivalent to the opposite of $\mathsf{CRing}$, say– this leads to many useful connections between geometric and algebraic concepts.

[3] if you'd like, you can say categories are precisely "monoidoids", and functors are "monoidoid homomorphisms", and I *do like* because "monoidoid" is a silly fun word and I like saying it

Now, sometimes we wish to talk about things which don't *respect* composition, but instead reverse it. These still get to be called functors, but we call those "**contravariant functors**" and (even if we *really*) think of them as "composition reversing functors $\mathcal{C} \to \mathcal{D}$) it'll end up being nice to think of these quite formally as functors $\mathcal{C}^{op} \to \mathcal{D}$– e.g. they "preserve reverse composition of arrows".
Now we know what a functor is, we should talk about some examples and basic properties thereof.

**Definition 1.4.** *A functor, $F : \mathcal{C} \to \mathcal{D}$, is said to be:*

- ***faithful*** *if the induced map $\mathsf{Hom}_{\mathcal{C}}(a, b) \to \mathsf{Hom}_{\mathcal{D}}(Fa, Fb)$ is injective for all $a, b \in \mathsf{obj}(\mathcal{C})$*
- ***full*** *if the induced map $\mathsf{Hom}_{\mathcal{C}}(a, b) \to \mathsf{Hom}_{\mathcal{D}}(Fa, Fb)$ is surjective for all $a, b \in \mathsf{obj}(\mathcal{C})$*
- ***bijective*** *if the induced map $\mathsf{Hom}_{\mathcal{C}}(a, b) \to \mathsf{Hom}_{\mathcal{D}}(Fa, Fb)$ is bijective for all $a, b \in \mathsf{obj}(\mathcal{C})$*

Now, in many contexts we want to compare two categories, talk about whether or not they're the "same". In such circumstances we *could* require that we have a pair of inverse functors– an "isomorphism of categories"– but the problem is that this is way too strong (to the point of coming down to "how strict are we about what sets we use to define our objects"). It is, for instance, perfectly valid linear algebra-wise to imagine all finite $\mathbb{R}$-vector spaces as just "vector spaces $\mathbb{R}^n$", but the two categories *aren't* isomorphic because one only has a $\mathbb{Z}^+$ worth of sets while the other has way more. The notion of "isomorphism" is thus the "wrong definition". For a better definition, rather than being "bijective on objects", we require a sort of weaker notion:

**Definition 1.5.** *A functor $F : \mathcal{C} \to \mathcal{D}$ is said to be **essentially surjective** if, for every $d \in \mathsf{obj}(\mathcal{D})$, there exists some $c \in \mathsf{obj}(\mathcal{C})$ such that $Fc \cong d$.*

That is to say, $F$ "reaches all of $\mathcal{D}$ (up to abstract isomorphism)" With this in mind, we can formulate the following:

**Definition 1.6.** *A functor $F : \mathcal{C} \to \mathcal{D}$ is said to be a **(weak) equivalence of categories** if $F$ is fully faithful and essentially surjective.*

This is to say: morphisms in $\mathcal{C}$ and $\mathcal{D}$ can be thought of interchangeably–so long as you're willing to transport along abstract isomorphisms in $\mathcal{D}$. Subscribing to the philosophy that "objects are but vessels for morphisms" (hey, I don't wanna bother with the yoneda lemma here), this tells us the two categories are "morally" the same. The reason for the parenthetical "weak" is more or less a "you can choose to be careful about the axiom of choice"– in particular, there's a natural notion of a "(homotopy) equivalence of categories"– cf. 1.9, which is nice and cool but requires a bit more terminology. With Axiom of choice these are the same, but without axiom of choice this notion is "weaker". Jason has said he often means "weak equivalence" when he says equivalence though (even though he's distinguished them in the past?), so mostly it's a "have the homotopic notion in mind morally, and the weak notion in mind for like 80% of practical examples."

**Definition 1.7.** *Given two functors, $F, G : \mathcal{C} \to \mathcal{D}$, a **natural transformation** $\eta_\bullet : F \to G$ is a collection of maps for each $a \in \mathsf{obj}(\mathcal{C})$, $\eta_a : Fa \to Ga$ such that for every $a \xrightarrow{\varphi} b$ in $\mathsf{Hom}(\mathcal{C})$ the following square commutes:*

$$\begin{tikzcd} Fa \arrow[r, "F\varphi"] \arrow[d, "\eta_a"] & Fb \arrow[d, "\eta_b"] \\ Ga \arrow[r, "G\varphi"'] & Gb \end{tikzcd}$$

**Remark 1.2.** *If the notation was not itself suggestive enough (and perhaps it wasn't at this high a level of abstraction, alas), a "natural transformation" is a good concept of "map between functors"– this should already be a somewhat familiar notion to those who know some representation theory: A representation of $G$ over $k$ (take $k = \mathbb{R}, \mathbb{C}$ for comfort if you like) is a sort of map (indeed, literally a functor) $G \to \mathsf{Vect}_k$, and an equivariant map between two representations is analogous to (indeed, literally is) a natural transformation between them. People also sometimes like to illustrate these with a "globular diagram"*

$$\begin{tikzcd} \mathcal{C} \arrow[r, bend left, "F"{name=U}] \arrow[r, bend right, "G"'{name=D}] & \mathcal{D} \arrow[from=U, to=D, Rightarrow, "\eta_\bullet"] \end{tikzcd}$$

*, which in particular should make you imagine a sort of "directed" homotopy from $F$ to $G$ (and indeed, one can kind of formulate homotopies of maps of topological spaces in this "maps between maps" language blabla)– in a bit, we'll see some concepts which are roughly "homotopy equivalences of categories", so this intuition is by no means off the mark.*

**Example 1.1.** *Denote* $(-)^\times : \mathsf{CRing} \to \mathsf{Group}$ *the functor that associates any commutative ring $R$ to its group of units. Denote* $GL_n(-) : \mathsf{CRing} \to \mathsf{Group}$ *the functor that associates any commutative ring $R$ to the group of invertible $n \times n$ matrices with coefficients in $R$. Then, the determinant map defines a natural transformation* $\det_\bullet : GL_n(-) \to (-)^\times$.

**Example 1.2.** *Denote* $(-)^\vee : {}_{\mathbb{R}}\mathsf{FinVect} \to {}_{\mathbb{R}}\mathsf{FinVect}$ *the functor that takes a vector space, $V$, to its dual* $V^\vee := \mathsf{Hom}_{\mathbb{R}}(V, \mathbb{R})$. *Now, for each finite dimensional vector space $V$, one can choose an isomorphism* $\eta_V : V \xrightarrow{\sim} V^\vee$ *by choosing a basis. Nonetheless, this can not be assembled into a natural transformation* $\mathrm{id}_{\mathcal{C}} \to (-)^\vee$*– intuitively because there is no way to make the choice of isomorphism "naturally" or "canonically"*

"why do we care about this?" well, to curry your question into a different question, we define another thing:

**Definition 1.8.** *A pair of functors $L : \mathcal{C} \to \mathcal{D}$, $R : \mathcal{D} \to \mathcal{C}$ are said to be an* ***adjoint pair*** *if any of the following equivalent definitions hold:*

- *(*$\mathsf{Hom}$ *bijection) For every* $c \in \mathsf{obj}(\mathcal{C}), d \in \mathsf{obj}(\mathcal{D})$, *there is a natural isomorphism* $\eta_{c,d} : \mathsf{Hom}_{\mathcal{D}}(Lc, d) \xrightarrow{\sim} \mathsf{Hom}_{\mathcal{C}}(c, Rd)$, *where "natural" is taken to mean "for any* $c \xrightarrow{\varphi} c' \in \mathsf{Hom}(\mathcal{C}), d \xrightarrow{\psi} d' \in \mathsf{Hom}(\mathcal{D})$, *the following commutes:*

$$\begin{tikzcd} \mathsf{Hom}_{\mathcal{C}}(Lc, d) \arrow[r, "\eta_{c,d}"] \arrow[d, "\mathsf{Hom}_{\mathcal{C}}(F\varphi,\psi)"'] & \mathsf{Hom}_{\mathcal{D}}(c, Rd) \arrow[d, "\mathsf{Hom}_{\mathcal{D}}(\varphi,R\psi)"] \\ \mathsf{Hom}_{\mathcal{C}}(Lc', d') \arrow[r, "\eta_{c',d'}"'] & \mathsf{Hom}_{\mathcal{D}}(c', Rd') \end{tikzcd}$$

- *(equational) We have natural transformations* $\eta : \mathrm{id}_{\mathcal{C}} \to R \circ L$ *(the "unit") and* $\varepsilon : L \circ R \to \mathrm{id}_{\mathcal{D}}$ *(the "counit") such that the following commutes:*

$$\begin{tikzcd} R \arrow[r, "\eta_{R(-)}"'] \arrow[rr, bend left, "\mathrm{id}_R"] & RLR \arrow[r, "R\varepsilon_{(-)}"'] & R \end{tikzcd}$$

$$\begin{tikzcd} L \arrow[r, "L\eta_{(-)}"] \arrow[rr, bend right, "\mathrm{id}_L"'] & LRL \arrow[r, "\varepsilon_{L(-)}"] & L \end{tikzcd}$$

*(in a quip, "$\varepsilon \circ \eta = \mathrm{id}$").*

- *(universal arrow for left adjoint) For every* $c \in \mathcal{C}$, *there exists* $Rc \in \mathsf{obj}(\mathcal{D})$ *and a map* $\varepsilon_c : LRc \to c$ *such that, for every* $d \in \mathsf{obj}(\mathcal{D})$ *and every* $\varphi : Ld \to c$, *there exists unique* $\tilde{\varphi} : d \to Rc$ *making the following diagram commute*

$$\begin{tikzcd} Ld \arrow[d, dashed, "\exists ! L\tilde{\varphi}"'] \arrow[dr, "\varphi"] & \\ LRc \arrow[r, "\varepsilon_c"'] & c \end{tikzcd}$$

- *(universal arrow for right adjoint) For every* $d \in \mathcal{D}$, *there exists* $Ld \in \mathsf{obj}(\mathcal{C})$ *and a map* $\eta_d : d \to RLd$ *such that, for every* $c \in \mathsf{obj}(\mathcal{C})$ *and every* $\psi : d \to Rc$, *there exists unique* $\tilde{\psi} : Ld \to c$ *making the* *following diagram commute*

$$\begin{tikzcd}
Rc & \\
RLd \arrow[u, dashed, "\exists! R\tilde{\psi}"] & d \arrow[l, "\eta_d"] \arrow[ul, "\psi"']
\end{tikzcd}$$

*In any of these cases, $L$ is said to be a* ***left adjoint*** *(to $R$), and $R$ is said to be a* ***right adjoint*** *(to $L$).*

**Remark 1.3.** *Each of these definitions gives a slightly different insight into what a "adjunction" really is, namely:*

**Remark 1.4.** *Technically, a proof that these definitions are equivalent is required. Practically, it's pure abstract nonsense and not particularly insightful. To see the proofs, look at the nlab pages on "adjunction" and "adjoint functors" , but you shouldn't need to remember them nearly so much as (the intuition behind) the definitions.*

Now that we have the requisite language, we can finally formulate a "(homotopy) equivalence of categories":

**Definition 1.9.** *Two categories, $\mathcal{C}, \mathcal{D}$ are said to be* ***equivalent*** *if there exists pair of functors $F : \mathcal{C} \to \mathcal{D}$ and $G : \mathcal{D} \to \mathcal{C}$ and a pair of natural isomorphisms $\eta : \mathrm{id}_{\mathcal{C}} \to F \circ G$ and $\varepsilon : G \circ F\, \mathrm{id}_{\mathcal{D}}$.*

**Remark 1.5.** *Any equivalence is an adjunction by the counit-unit definition above, and indeed: an equivalence is precisely an adjunction for which the counit and unit transformations are iso.*

## 1.1 Yoneda Lemma

**Definition 1.10.** *Let $\mathcal{C}$, $\mathcal{D}$ be categories. Then, we denote $\mathsf{Fun}(\mathcal{C}, \mathcal{D})$ the category of (covariant) functors from $\mathcal{C} \to \mathcal{D}$, where the morphisms are natural transformations.*

The above definition actually *doesn't* always "make sense" in that we required categories to have a *set* of morphisms between any two objects, and yet this definition (at least taken as literally as possible) does not always meet this criterion. We shall ignore such issues for now.

**Definition 1.11.** *Let $\mathcal{C}$ be a category. We denote by $\text{よ}_\bullet : \mathcal{C} \to \mathsf{Fun}(\mathcal{C}, \mathsf{Set})$ the functor $c \mapsto \text{よ}_c := \mathsf{Hom}_{\mathcal{C}}(c, -)$, and we denote $\text{よ}^\bullet : \mathcal{C}^{op} \to \mathsf{Fun}(\mathcal{C}, \mathsf{Set})$ the functor $c \mapsto \text{よ}^c := \mathsf{Hom}_{\mathcal{C}}(-, c)$. These are called the* ***covariant and contravariant Yoneda functors*** *respectively (with the symbol used being the hiragana character "yo" for Yoneda). The notation $h_\bullet$ or $h^\bullet$ is also used.*

**Theorem 1.1.** *(Yoneda Lemma) There is a natural isomorphism between the functors*

$$\begin{gathered}
\mathsf{eval} : \mathcal{C} \times \mathsf{Fun}(\mathcal{C}, \mathsf{Set}) \to \mathsf{Set} \\
(c, F) \mapsto Fc \\
(\varphi : c \to c', \eta : F \to G) \mapsto \eta_{c'} \circ F\varphi
\end{gathered}$$

*and*

$$\begin{gathered}
\mathsf{Hom}_{\mathsf{Fun}(\mathcal{C}, \mathsf{Set})}(\text{よ}_\bullet, -) : \mathcal{C} \times \mathsf{Fun}(\mathcal{C}, \mathsf{Set}) \to \mathsf{Set} \\
(c, F) \mapsto \mathsf{Hom}_{\mathsf{Fun}(\mathcal{C}, \mathsf{Set})}(\text{よ}_c, F) \\
(\varphi : c \to c', \eta : F \to G) \mapsto \mathsf{Hom}_{\mathsf{Fun}(\mathcal{C}, \mathsf{Set})}(\text{よ}_\varphi, \eta).
\end{gathered}$$

*(as well as a dual one for the contravariant yoneda map). This is to say, we have natural bijections*

$$\mathsf{Hom}_{\mathsf{Fun}(\mathcal{C}, \mathsf{Set})}(\text{よ}_c, F) \cong Fc$$

*and*

$$\mathsf{Hom}_{\mathsf{Fun}(\mathcal{C}^{op}, \mathsf{Set})}(\text{よ}^c, F) \cong Fc.$$

*Proof.* For all $\varphi : c \to c', F : \mathcal{C} \to \mathsf{Set}$, we have a natural transformation $\vartheta : \text{よ}_c \to F$ given by the commutative diagram

$$\begin{tikzcd}
\text{よ}_c(c) = \mathsf{Hom}_{\mathcal{C}}(c,c) \arrow[rrr, "\mathsf{Hom}_{\mathcal{C}}(c,\varphi) =: \text{よ}_c\varphi"] \arrow[ddd, "\vartheta_c"] & & & \text{よ}_c(c') = \mathsf{Hom}_{\mathcal{C}}(c,c') \arrow[ddd, "\vartheta_{c'}"] \\
& \mathrm{id}_c \arrow[r, mapsto] \arrow[d, mapsto] & \varphi \arrow[d, mapsto] & \\
& * \arrow[r, mapsto] & F\varphi(*) = \vartheta_{c'}\varphi & \\
Fc \arrow[rrr, "F\varphi"] & & & Fc'
\end{tikzcd}$$

Any natural transformation $\vartheta$ is determined by the choice of $\vartheta_c(\mathrm{id}_c) = * \in F(A)$, and any choice of $*$ defines such a natural transformation.
An analogous diagram proves the analogous result for $\text{よ}^\bullet$. $\square$

Alright, so that was some very abstract nonsense. Key upshot (for now):

**Corollary 1.1.** *Two $\mathcal{C}$-objects, $c, c'$, are isomorphic if and only if their* Hom *functors (contravariant or covariant) are isomorphic.*

*Proof.* By the above, we have

$$\mathsf{Hom}_{\mathsf{Fun}(\mathcal{C},\mathsf{Set})}(\text{よ}_c, \text{よ}_{c'}) \cong \mathsf{Hom}_{\mathcal{C}}(c', c)$$

and

$$\mathsf{Hom}_{\mathsf{Fun}(\mathcal{C}^{op},\mathsf{Set})}(\text{よ}^c, \text{よ}^{c'}) \cong \mathsf{Hom}_{\mathcal{C}}(c, c').$$

$\square$

**Remark 1.6.** *Because of the above result, the functors $\text{よ}^\bullet$ and $\text{よ}_\bullet$ are often called the "**(resp. contravariant, covariant) Yoneda embedding**", as we see that these functors are fully faithful (and isomorphism reflecting).*[4]

Philosophically, here's the point: "objects in $\mathcal{C}$ are determined by morphisms to (and/or from) them". This is, more or less, a far *far* reaching generalization of Cayley's theorem in group theory ("every group encodes the symmetries of some object").

## 1.2 (Co)Limits

**Definition 1.12.** *In a category, $\mathcal{C}$, a "**diagram**" is a functor $F : \mathcal{I} \to \mathcal{C}$ where $\mathcal{I}$ is small (and thought of as a "index category").*

The term "index category" here is taken in analogy to that of "index set", e.g. for a product/direct sum.

**Example 1.3.** *Take $\mathcal{I}$ to be the category given by the following digraph (with identity maps implicit):*

$$1 \xleftarrow{i} 3 \xrightarrow{j} 2$$

*Take $\mathcal{C} = \mathsf{CRing}$. Then, the functor such that $F(1) = \mathbb{Z}[x]$, $F(2) = \mathbb{Z}/\nvDash\mathbb{Z}, F(3) = \mathbb{Z}$ with $F(i), F(j)$ taken to be the natural inclusions gives a diagram, visualized as*

$$\mathbb{Z}[x] \longleftarrow \mathbb{Z} \xrightarrow{j} \mathbb{Z}/2\mathbb{Z}$$

[4] Sometimes, it is considered useful to view certain nice contravariant functors from certain nice ("geometric/topological") categories as "generalized spaces" ("sheaves and/or algebraic spaces") which arise from gluing the objects/spaces in the base category. In this view, a "sheaf" which is isomorphic to $\text{よ}^c$ is said to be "represented by $c$"– such matters are quite relevant to the study of moduli spaces/moduli problems.

**Example 1.4.** *Take $\mathcal{I}$ to be the category given by the following infinite digraph (with identity maps implicit):*

$$1 \xrightarrow{i_{12}} 2 \xrightarrow{i_{23}} 3 \xrightarrow{i_{34}} \dots$$

*Take $\mathcal{C} = \mathsf{CRing}$. Then, the functor such that, for every positive integer $n$, we take $F(n) = \mathbb{Z}/2^n\mathbb{Z}$ and $F(i_{n(n+1)})$ is taken to be the natural inclusion $\mathbb{Z}/2^n\mathbb{Z} \to \mathbb{Z}/2^{n+1}\mathbb{Z}$ is a diagram, visualized as*

$$\mathbb{Z}/2\mathbb{Z} \longrightarrow \mathbb{Z}/2^2\mathbb{Z} \longrightarrow \mathbb{Z}/2^3\mathbb{Z} \longrightarrow \dots$$

**Example 1.5.** *Take $\mathcal{I}$ to be the category given by the following digraph (with identity maps implicit):*

$$1 \qquad\qquad 2$$

*Take $\mathcal{C} = \mathsf{CRing}$. Then, the functor with $F(1) = \mathbb{R}$ abd $F(2) = \mathbb{C}$ gives a diagram, visualized as:*

$$\mathbb{R} \qquad\qquad \mathbb{C}$$

**Definition 1.13.** *Let $F : \mathcal{I} \to \mathcal{C}$ be a diagram. A "**cone over**" (resp. "**cocone under**") is a $\mathcal{C}$-object $c$, together with arrows $c \xrightarrow{\pi_n} Fn$ (resp. $c \xleftarrow{\iota_n} Fn$) for each $n \in \mathsf{obj}(\mathcal{I})$, such that for every $n \xrightarrow{i_{nm}} m \in \mathsf{Hom}(\mathcal{I})$ we have that the following diagram on the left (resp. on the right) commutes:*

$$\begin{tikzcd} & c \arrow[dl, bend right, "\pi_n"'] \arrow[dr, bend left, "\pi_m"] & \\ Fn \arrow[rr, "F(i_{nm})"] & & Fm \end{tikzcd} \qquad \begin{tikzcd} Fn \arrow[rr, "F(i_{nm})"] \arrow[dr, bend right, "\iota_n"'] & & Fm \arrow[dl, bend left, "\iota_m"] \\ & c & \end{tikzcd}$$

**Definition 1.14.** *Let $F : \mathcal{I} \to \mathcal{C}$ be a diagram. A **limit** (resp. **colimit**) is a terminal cone over $F$ (resp. initial cocone under $F$)– e.g, an object $c$ equipped with maps $c \xrightarrow{\pi_n} Fn$ (resp. $c \xleftarrow{\iota_n} Fn$) such that, for any other object $c'$ equipped with maps $c' \xrightarrow{\pi'_n} Fn$ (resp. $c' \xleftarrow{\iota'_n} Fn$), there exists a unique map $\pi : c' \to c$ (resp $\iota : c \to c'$) such that the following diagram on the left (resp. on the right) commutes:*

$$\begin{tikzcd} & c' \arrow[d, dashed, "\exists!\pi"] \arrow[ddl, bend right, "\pi'_n"'] \arrow[ddr, bend left, "\pi'_m"] & \\ & c \arrow[dl, bend right, "\pi_n"'] \arrow[dr, bend left, "\pi_m"] & \\ Fn \arrow[rr, "F(i_{nm})"] & & Fm \end{tikzcd} \qquad \begin{tikzcd} Fn \arrow[rr, "F(i_{nm})"] \arrow[dr, bend right, "\iota_n"'] \arrow[ddr, bend right, "\iota'_n"'] & & Fm \arrow[dl, bend left, "\iota_m"] \arrow[ddl, bend left, "\iota'_m"] \\ & c \arrow[d, dashed, "\exists!\iota"] & \\ & c' & \end{tikzcd}$$

**Example 1.6.** *Let $F : \mathcal{I} \to \mathcal{C}$ be as in example 1.3. Then, the colimit over the diagram is given by the tensor product as (commutative) $\mathbb{Z}$-algebras, $\mathbb{Z}[x] \otimes_{\mathbb{Z}} \mathbb{Z}/2\mathbb{Z} \cong \mathbb{Z}/2\mathbb{Z}[x]$:*

$$\begin{tikzcd} \mathbb{Z}[x] \otimes_{\mathbb{Z}} \mathbb{Z}/2\mathbb{Z} & \mathbb{Z}[x] \arrow[l] \\ \mathbb{Z}/2\mathbb{Z} \arrow[u] & \mathbb{Z} \arrow[l] \arrow[u] \end{tikzcd}$$

*This is also called a **pushout** (its dual is called a **pullback** or a **fibre product**).*

**Example 1.7.** *Let $F : \mathcal{I} \to \mathcal{C}$ be as in example 1.4. Then, the limit over the diagram is given by the 2-adic integers $\mathbb{Z}_2$ (look them up on wikipedia if you haven't seen them before, they're useful for number theory etc.), together with the natural projections:*

$$\begin{tikzcd} & \mathbb{Z}_2 & & \\ \mathbb{Z}/2\mathbb{Z} \arrow[ur] \arrow[r] & \mathbb{Z}/2^2\mathbb{Z} \arrow[u] \arrow[r] & \mathbb{Z}/2^3\mathbb{Z} \arrow[ul] \arrow[r] & \dots \end{tikzcd}$$

*This is also called a **projective/inverse limit** (its dual is called a **direct/inductive limit**).*

**Example 1.8.** *Let $F : \mathcal{I} \to \mathcal{C}$ be as in example 1.5. Then, the limit over the diagram is given by the product as rings, and the colimit over the diagram is given by the tensor product as $\mathbb{Z}$-algebras (note $\mathbb{Z}$ the initial object of* $\mathsf{CRing}$*, so one has "invisible" maps $\mathbb{Z} \to \mathbb{R}$ and $\mathbb{Z} \to \mathbb{C}$):*

$$\begin{tikzcd} & \mathbb{R} \times \mathbb{C} \arrow[dl, "\pi_{\mathbb{R}}"'] \arrow[dr, "\pi_2"] & \\ \mathbb{R} & & \mathbb{C} \end{tikzcd} \qquad \begin{tikzcd} \mathbb{R} \arrow[dr, "\iota_1"'] & & \mathbb{C} \arrow[dl, "\iota_2"] \\ & \mathbb{R} \otimes_{\mathbb{Z}} \mathbb{C} & \end{tikzcd}$$

*The left limit is called a **product**, the right colimit is called a **coproduct** (though indeed, coproducts in* $\mathsf{CRing}$ *can be thought of as pushouts under $\mathbb{Z}$).*

Idea: colimits usually look like disjoint unions modulo identifying some things, limits usually look like subcollections of products. Google examples of (co)products, pushouts, and pullbacks in categories you know about (including $\mathsf{Top}$ by the way) to feel a bit more comfortable with the terminology. Important note: in module categories, finite (but not infinite!) products and coproducts coincide. "coproducts" bear the name of "direct sum" more generally, but finite direct sums *are* products as well (which is part of what makes "abelian categories" so nice).

**Theorem 1.2.** *Suppose $F : \mathcal{C} \to \mathcal{D}$ is a functor. Then:*

- *If $F$ is a left adjoint (e.g., has a right adjoint), then $F$ preserves colimits.*
- *If $F$ is a right adjoint (e.g., has a left adjoint), then $F$ preserves limits.*

This has a proof by pure formal nonsense given by unraveling the definitions. This is also an incredibly useful fact from which many other properties follow. If it helps to remember, one has the mnemonics "LAPC [left adjoints preserve colimits]" and "RAPL [right adjoints preserve limits]."

**Proposition 1.1.** *Let $F : \mathcal{I} \to \mathcal{C}$ be a diagram. Then, a limit of $F$ (resp. a colimit of $F$) is precisely a $\mathcal{C}$-object $\lim F$ (resp. $\operatorname{colim} F$) such that, for all $c \in \mathsf{obj}(\mathcal{C})$ we have a natural bijection*

$$\mathsf{Hom}_{\mathcal{C}}(c, \lim F) \cong \mathsf{Hom}_{\mathsf{Fun}(\mathcal{I},\mathcal{C})}(\mathsf{const}_c, F)$$

*or respectively a natural bijection*

$$\mathsf{Hom}_{\mathcal{C}}(\operatorname{colim} F, c) \cong \mathsf{Hom}_{\mathsf{Fun}(\mathcal{I},\mathcal{C})}(F, \mathsf{const}_c),$$

*where by $\mathsf{Fun}(\mathcal{I}, \mathcal{C})$ we mean the category of functors and natural transformations (ignore set-theoretic issues here), and by $\mathsf{const}_c$ we mean the functor that takes every object to $c$ and every morphism to $\mathrm{id}_c$.*
*Rephrasing: $\lim_{\mathcal{I}}$ and $\operatorname{colim}_{\mathcal{I}}$ are right (resp. left) adjoint to the functor $\mathsf{const} : \mathcal{C} \to \mathsf{Fun}(\mathcal{I}, \mathcal{C})$.*

This above proposition, modulo set-theoretic issues w.r.t. defining a functor category, is a valid alternative definition of "limit/colimit", and can lead one to various useful ideas (e.g., the 2-category of categories, viewing $\lim / \operatorname{colim}$ as functors on their own right, left/right Kan extensions, etc.). It is a fine exercise to try and unravel this definition and see how the universal property (w.r.t. cones/cocones) is equivalent to this statement, but we would suggest not overemphasizing this definition if any of these concepts are new to the reader.

# 2 Tensor products (even over potentially non-commutative rings) and more miscellaneous module theory

**Definition 2.1.** ***R*-balanced maps** *Let $R$ be a ring, $M \in \mathsf{Mod}_R, N \in {}_R\mathsf{Mod}$. Then, an $R$ balanced map $\varphi : M \times N \to A$ an abelian group is a biadditive map (e.g., $\mathbb{Z}$-bilinear map) such that, for all $r \in R$, $\varphi(mr, n) = \varphi(m, rn)$ for all $m \in M, n \in N$.*

**Definition 2.2.** ***tensor product*** *Let $M \in {}_S\mathsf{Mod}_R, N \in {}_R\mathsf{Mod}_{S'}$ for a ring $R$. Then, the tensor product $M \otimes_R N$ is an abelian group (in particular, an $S$-$S'$-bimodule) together with a map $\otimes_R : M \times N \to M \otimes_R N$ such that, for any abelian group $A$ together with an $R$-balanced map $\varphi : M \times N \to A$, there exists unique abelian group homomorphism $\tilde{\varphi} : M \otimes_R N \to A$ making the following diagram commute:*

$$\begin{tikzcd} M \times N \arrow[r, "\otimes"] \arrow[dr, "\varphi"'] & M \otimes_R N \arrow[d, dashed, "\exists! \tilde{\varphi}"] \\ & A \end{tikzcd}$$

*. If $R$ is commutative, then $M \otimes_R N$ admits the natural structure of an $R$-module, and we can instead consider the universal $R$-module such that any $R$-bilinear map $M \times N \to A$ factors through an $R$-module homomorphism $M \otimes_R N \to A$.*

Note: while we take bimodules above, we can simply take $S$, $S'$, or $R$ to be $\mathbb{Z}$ and recover "one-sided" modules. That's a universal property or what have you– really, it's the same philosophy as viewing the "tensor product of $V$ and $W$" as "the multilinear maps from $V \times W$", we're just being cautious about commutativity and being a smidgeon more careful since not every module is free here. Anyways, "how/why does this universal object actually exist?", well, we just take a super big free module and the mod out by what it means to be $R$-balanced, aka by what it means to be (non-commutatively) multilinear. We give the proof of existence as follows:

**Lemma 2.1.** *)(tensor products exist)*

The tensor product has, itself, a deep relation to $\mathsf{Hom}$-functors– namely, we have the *tensor-hom adjunction*:

**Theorem 2.1.**

**Definition 2.3.** *Let $R$ be a ring. An **$R$-algebra** is a ring, $S$, together with a ring homomorphism $R \to S$ (the "structure map").*

**Definition 2.4.** *Let $R \to S$, $R \to S'$ be two $R$-algebras. A **(homo)morphism of $R$-algebras** from $S \to S'$ is a ring homomorphism $\varphi : S \to S'$ which commutes with the structure maps, e.g. making the following diagram commute:*

$$\begin{tikzcd} & R \arrow[dl] \arrow[dr] & \\ S \arrow[rr, "\varphi"] & & S' \end{tikzcd}$$

**Definition 2.5.** *Let $R$ be a commutative ring, let $R \to S$ and $R \to S'$ be $R$-algebras. Then, the tensor product of $R$-algebras is the colimit in $\mathsf{CRing}$ over the diagram*

$$S' \leftarrow R \to S,$$

*e.g. the pushout in $\mathsf{CRing}$ (or, as one can verify, the coproduct in ${}_R\mathsf{CAlg}$).*

**Remark 2.1.** *Note that the tensor product of $R$-algebras can also be taken in steps. Namely, any $R$-algebra $R \to S$ admits the natural structure of an $R$-module by left/right multiplication. Then, for two $R$-algebras $S$ and $S'$, one can take their tensor product as $R$-modules, and then endow $S \otimes_R S'$ with the structure of an ring by defining $(s_1 \otimes s_1')(s_2 \otimes s_2') := s_1 s_2 \otimes s_1' s_2'$, which one can check gives an $R$-algebra in the natural way.*

**Definition 2.6.** *Let $R$ be a commutative ring, and $S \subset R$ a multiplicatively closed subset. The* ***localization****, $S^{-1}R$ is taken to be a (commutative) $R$-algebra $R \to S^{-1}R$ such that, for any ring homomorphism $R \to T$ taking every element of $S$ to a unit, there exists a unique ring homomorphism $S^{-1}R \to T$ making the following diagram commute:*

$$\begin{tikzcd} R \arrow[r] \arrow[dr] & S^{-1}R \arrow[d, dashed, "\exists!"] \\ & T \end{tikzcd}$$

This is to say, $S^{-1}R$ is the "smallest" (e.g., initial) $R$-algebra for which every element of $S$ is a unit. As a universal property, this requires a construction/proof of existence– this one justifies the common name "ring of fractions" this ring.

**Theorem 2.2.** *(Construction of localizations) Localizations exist.*

*Proof.* Let $S$ be a multiplicative subset of $R$ commutative. Define, on $R \times S$, an equivalence relation

$$(a, b) \sim (c, d)$$

if there exists $s \in S$ such that

$$s(ac - bd) = 0.$$

As this forms an ideal, we may take $S^{-1}R$ to be the ring of equivalence classes under this relation; we denote the equivalence class of $(r, s)$ as either $\frac{r}{s}$ or $s^{-1}r$. Then, the map $r \mapsto \frac{r}{1} : R \to S^{-1}R$ makes $S^{-1}R$ initial among all $R$-algebras for which $S$ is taken to units by the structure map. $\square$

The relation in question should be taken to be exactly that of fraction addition working "as it should". The use of an extra $s \in S$ in that definition is mostly just to fudge things a little so that we don't need to make an extra case for the multiplicative set having zero divisors (but, for $S$ to be a multiplicative set containing a zero divisor, it must also contain 0 and hence the localization must be the zero ring anyways.)
There is some special notation in the following cases:

**Definition 2.7.** *If $f \in R$, then we denote $R_f := \{f, f^2, f^3, \dots\}^{-1}R$ the "**localization away from** $f$". If $\mathfrak{p}$ is a prime ideal of $R$, then we denote $R_{\mathfrak{p}} := (R - \mathfrak{p})^{-1}R$ the "**localization at** $\mathfrak{p}$."*[5]

[5] This terminology comes from algebraic geometry, where prime ideals (or, intuitively, maximal ideals) are thought of as "points" of a topological space. Imagining $R$ as the ring of continuous functions on a compact Hausdorff space, $\mathfrak{p}$ as a point of said space, then we have the localization *at* $\mathfrak{p}$ arising as the "ring of germs of functions at $\mathfrak{p}$". The terminology for localization *away from* an element is analogous: the localization at $f \in R$ is precisely the localization (intuitively, restriction to) all prime ideals ("points") not containing $f$ (that is, where $f$ does not vanish).

# 3 (co)homological algebra

**Definition 3.1.** *If $F : {}_R\mathsf{Mod} \to {}_S\mathsf{Mod}$ is a functor of module categories (it can be different sided module categories too, it doesn't matter), $F$ is said to be additive if, given any two $\varphi, \psi \in \mathsf{Hom}_R(M, N)$, we have $F(\varphi + \psi) = F(\varphi) + F(\psi)$– that is, on each $\mathsf{Hom}$ it's a homomorphism of additive groups.*

I'm really only mentioning the word "additive" here so that I can say "additive" later on without needing to refer you to the stuff on abelian categories (at least, not yet).

**Definition 3.2.** *An $R$ module, $P$, is said to be* ***projective*** *if any of the following equivalent statements are true:*

- *For any surjective module homomorphism $N \twoheadrightarrow M$, any module homomorphism $P \to M$ admits a (not-necessarily unique) lift $P \to N$ such that the following diagram commutes:*

$$\begin{tikzcd} & N \arrow[d] \\ P \arrow[ur, dashed] \arrow[r] & M \end{tikzcd}$$

- *Any short exact sequence*

$$0 \to M \to N \to P \to 0$$

  *splits.*

- *The (covariant) functor $\mathsf{Hom}_R(P, -)$ is exact (note left-exactness is always true for $\mathsf{Hom}$ functors)*

- *$P$ is a direct summand of a free module*[6]

**Remark 3.1.** *Any free module is projective. Furthermore, the converse is occasionally (but not always!) true: a projective module is always free over a PID (basic fact), and a projective module over a local ring is always free (straightforward to prove for finitely generated modules via Nakayama's lemma from commutative algebra, but a difficult fact in general– keep the general result in mind only as a restriction for coming up with counterexamples).*

**Definition 3.3.** *An $R$-module, $F$, is said to be* ***flat*** *if any of the following equivalent conditions are true:*

- *The functor $F \otimes_R - \cong - \otimes_R F$ is exact (note right-exactness is always true for tensoring)*

- *The functor $F \otimes_R -$ takes injective maps to injective maps.*

- *(equational criterion for flatness) ["relations in $F$ arise from relations in $R$"] For any $R$-module homomorphism $\varphi : R^n \to F$, and any $K$ a finitely generated submodule of $\ker \varphi$, $\varphi$ factors through a map $\psi : R^n \to F'$ for some free*[7] *$R$-module$F'$ so that $K \subset \ker \psi$, e.g. such that the following diagram commutes:*

$$\begin{tikzcd} & & & F' \arrow[dr, dashed] & \\ K \arrow[r, hook, "f.g."] \arrow[dr, hook, dashed] & \ker\varphi \arrow[r, hook] & R^n \arrow[ur, dashed, "\exists\psi"] \arrow[rr, "\varphi"] & & F \\ & \ker\psi \arrow[ur, hook, dashed] & & & \end{tikzcd}$$

[6] Note: not every abelian category has a good notion of "free object", so this particular description is distinct for categories of modules.

[7] Note: not every abelian category has a good notion of "free object", so this particular description is distinct for categories of modules.

- *(translating the above into a concrete statement about relations– thank you wikipedia) For any R-linear relation*
$$\sum_{i=1}^{n} r_i x_i = 0$$
*with $r_i \in R$, $m_i \in F$, there exist some ("change of basis/generator") elements $y_j \in F, a_{ij} \in R$ such that*
$$\forall j,\ \sum_{i=1}^{n} r_i a_{ij} = 0$$
*and*
$$\forall i,\ x_i = \sum_{j=1}^{m} a_{ij} y_j.$$

**Remark 3.2.** *Any projective module is necessarily flat. The converse is ("clearly") true for finitely presented flat modules (hence for finitely generated flat modules over Noetherian rings).*

There is yet another equivalent condition (which is, to my understanding, never taken as a definition):

**Theorem 3.1.** *$M$ is a flat $R$ module if and only if, for every (finitely generated) ideal $I$ of $R$, $M \otimes I \to M \otimes R$ is injective.*

For a particularly elementary (if long) proof, see Dummit and Foote exercise 10.5.25 (more or less– an $R$). When we define (left)-derived functors, we will have another fancier restatement of the above available to us:

**Theorem 3.2** (ideal-theoretic criterion for flatness)**.** *$M$ is a flat $R$-module if and only if*
$$\mathsf{Tor}_1^R(M, R/I) = 0$$
*for all (finitely generated) ideals $I$ of $R$.*

**Definition 3.4.** *An R-module, $I$, is said to be **injective** if any of the following equivalent statements are true:*

- *For any module homomorphism $M \to I$, any injective module homomorphism $M \otimes N$ admits an extension $N \to I$ ("along $M \to I$")such that the following diagram commutes:*
$$\begin{tikzcd} I & M \arrow[l] \arrow[d, hook] \\ & N \arrow[ul, dashed] \end{tikzcd}$$
- *Any short exact sequence*
$$0 \to I \to M \to N \to 0$$
*splits.*
- *The (contravariant) functor $\mathsf{Hom}_R(-, I)$ is exact (note left-exactness [interpreted appropriately for variance] is always true for $\mathsf{Hom}$ functors)*

Note that projective modules are "dual" to injective modules and vice versa. Here is a useful criterion for checking injectivity:

**Theorem 3.3.** *(Baer's Criterion) Let $I$ be an $R$-module. Then, $I$ is injective if and only if any homomorphism $J \to I$ from $J$ a submodule/ideal of $R$ admits an extension to a homomorphism $R \to I$.*

As an immediate corollary, we have the following:

**Corollary 3.1.** *Any abelian group $A$ is injective (as a $\mathbb{Z}$-module) if and only if it is divisible (e.g., the map $a \mapsto na$ is a surjection for all non-zero $n \in \mathbb{Z}$).*

## 3.1 Ext and Tor

Now that we know that Tor is, we can provide a quick proof of theorem 3.2:

*Proof of theorem 3.2.* The forward implication is immediate (from Tor being balanced). For the reverse implication, We follow 24.3.4 in Vakil's FOAG.
Note that $M$ is flat if and only if $\mathsf{Tor}_1^R(M, N) \cong \mathsf{Tor}_1^R(N, M) = 0$ for all $N$ (in which case the LES of Tor gives a SES).
Now, if $N$ is finitely generated take $\{r_1, \dots, r_n\}$ a set of generators. We have $A/\operatorname{Ann}(r_n) \cong Ar_n$ a submodule of $N$, hence we have a SES

$$0 \to A/\operatorname{Ann}(r_n) \to N \to N/\langle r_n \rangle \to 0$$

a SES where $N/\langle r_n \rangle$ is generated by $\langle r_1, \dots, r_{n-1}. \rangle$. Taking Tor and inducting on $n$ we get the desired result for finitely generated modules, since the base case $n = 1$ gives $N \cong R/\operatorname{Ann}(r_1)$.
In general, a module $N$ is a union of its finitely generated submodules, hence a *filtered colimit* (the filtering given by set-theoretic inclusion) of its f.g. submodules. By general abstract nonsense, filtered colimits are exact functors (exercise 1.5.L of Vakil) hence commute with homology by theorem 4.2, hence $\mathsf{Tor}_1^R(M, N) = 0$. $\square$

# 4 Generalities on "Abelian Categories"

**Remark 4.1.** *Jason might require us to technically be able to unravel the definition, but, in practice "abelian category" just means "category which we can think of like a category of left/right R-modules". This is possible to formalize by the 4.1Freyd-Mitchell Embedding Theorem, which is technically useful. It's also complete abstract nonsense, and for every almost every single thing we talk about, you can use the words "abelian category" interchangeably with "category of R-modules" (or, if you really want, "category of chain complexes of R-modules"), so just think in terms of whatever's easiest.*

**Theorem 4.1.** *(Freyd-Mitchell Embedding) Let $\mathcal{A}$ be a (small) abelian category. Then, there is a ring $R$ (not necessarily commutative) and a full, faithful, exact (hence additive) functor $i : \mathcal{A} \otimes {}_R\mathsf{Mod}$ the category of left R-modules.*

You do *not* need to memorize this (nor will we see a proof), but I feel it necessary to mention it as a name to look up/be referred to.

**Definition 4.1.** *A* $\delta$***-functor***

**Definition 4.2.** *A $\delta$-functor, $H^*$* ***universal*** *if*

**Theorem 4.2.** *(FOAG 1.5.I, "the FHHF theorem") Suppose $F : \mathcal{A} \to \mathcal{B}$ is a functor of abelian categories, and $C^\bullet$ an $\mathcal{A}$-complex.*

- *If $F$ is right exact, there exists a natural morphism $FH^\bullet \to H^\bullet F$*
- *If $F$ is left exact, there is a natural morphism $H^\bullet F \to FH^\bullet$ (e.g., $FH^\bullet \leftarrow H^\bullet F$)*
- *If $F$ is exact, then there is a natural isomorphism $FH^\bullet \xrightarrow{\sim} H^\bullet F$.*

*Proof.* □

This is abstract nonsense, but generally quite useful abstract nonsense (in that it gives very quick/easy proofs, e.g. that localization commutes with homology etc).[8]

[8]For those so inclined, one can google the definition of "filtered category" (a categorical generalization of "poset of set-theoretic inclusions of subsets") to make sense of a "filtered colimit"- Then, putting in a bit of work, one can show that taking filtered colimits gives an exact functor *on module categories*. (This is not true in general, cf. Grothendieck's AB5 axiom for "nice(r) abelian categories.")